

Datamatics Agentic AI Assessment

Compliance Validator Agent Challenge

Duration: 3-4 Days | **Difficulty:** Advanced

Background

You have been hired by FinanceGuard Solutions, a mid-sized accounts payable automation company. Their existing rule-based compliance system is failing at scale - it catches only 67% of compliance violations and generates 34% false positives.

Your task is to build an Agentic AI Compliance Validator that can intelligently validate invoices against Indian GST/TDS regulations with human-like judgment.

The Challenge

Build a multi-agent system that performs 58-point compliance validation on invoices. Your system must:

1. Process invoices in multiple formats (PDF, images, JSON, CSV)
2. Validate against GST and TDS regulations
3. Handle ambiguous cases with reasoning
4. Generate actionable compliance reports
5. Learn from corrections (bonus)

What You're Given

The following data files are provided:

- invoices/ - 150 test invoices (mixed formats)
- vendor_registry.json - Registered vendor details
- gst_rates_schedule.csv - GST rate mappings
- hsn_sac_codes.json - HSN/SAC code master
- tds_sections.json - TDS applicability rules
- company_policy.yaml - Internal approval thresholds
- historical_decisions.jsonl - Past compliance decisions (some incorrect!)

The 58-Point Validation Framework

Your agent must validate across these categories:

Category A: Document Authenticity (8 points)

ID	Check	Complexity
A1	Invoice number format validation	Low
A2	Duplicate invoice detection (across vendors)	Medium
A3	Sequential invoice number gap analysis	High
A4	Digital signature verification	Medium
A5	Invoice date vs document metadata date	High
A6	Seller details match vendor registry	Medium
A7	Buyer GSTIN matches company records	Low
A8	Invoice tampering detection (for images/PDFs)	High

Category B: GST Compliance (18 points)

ID	Check	Complexity
B1	GSTIN format validation (15-char alphanumeric)	Low
B2	GSTIN active status verification	Medium
B3	State code in GSTIN matches address	Medium
B4	HSN/SAC code validity	Low
B5	HSN code matches product description	High
B6	GST rate matches HSN/SAC code	Medium
B7	CGST + SGST = IGST rate validation	Low
B8	Inter-state vs Intra-state tax type	Medium
B9	Place of supply determination	High
B10	Reverse charge mechanism applicability	High
B11	Composition scheme validation	Medium
B12	E-invoice/IRN validation for B2B > 5Cr	High
B13	QR code presence and validity	Medium
B14	IRN hash verification	High
B15	Invoice value threshold for E-invoice	Medium

B16	Export invoice compliance (LUT/Bond)	High
B17	SEZ supply validation	High
B18	Input tax credit eligibility	High

Category C: Arithmetic & Calculation (10 points)

ID	Check	Complexity
C1	Line item quantity x rate = amount	Low
C2	Subtotal matches sum of line items	Low
C3	Tax calculation accuracy ($\pm ₹1$ tolerance)	Medium
C4	Discount application correctness	Medium
C5	Rounding rules compliance	Medium
C6	Multi-currency conversion accuracy	High
C7	Advance adjustment reconciliation	High
C8	Credit note/Debit note linkage	High
C9	Partial payment calculations	Medium
C10	TDS amount calculation	Medium

Category D: TDS Compliance (12 points)

ID	Check	Complexity
D1	TDS applicability based on vendor type	Medium
D2	Section determination (194C/194J/194H etc.)	High
D3	TDS rate based on PAN availability	Medium
D4	Lower deduction certificate validation	High
D5	Threshold limit for TDS applicability	Medium
D6	Aggregate payment threshold tracking	High
D7	TDS on GST component handling	High
D8	Non-resident TDS rules	High
D9	TAN validation	Low
D10	Section 206AB higher rate applicability	High
D11	TDS certificate availability	Medium
D12	Quarterly reconciliation flags	High

Category E: Policy & Business Rules (10 points)

ID	Check	Complexity
E1	Invoice within PO amount tolerance ($\pm 5\%$)	Medium
E2	Invoice date within contract period	Low
E3	Approved vendor list check	Low
E4	Spending category authorization	Medium
E5	Budget availability check	High
E6	Approval hierarchy based on amount	Medium
E7	Duplicate payment prevention	High
E8	Aging beyond payment terms	Low
E9	Early payment discount eligibility	Medium
E10	Retrospective invoice rules	High

Critical Constraints & Edge Cases

Constraint 1: Conflicting Regulations

Some invoices will have scenarios where GST and TDS rules conflict or are ambiguous. Your agent must:

- Identify the conflict
- Reason through both interpretations
- Make a recommendation with confidence score
- Flag for human review when confidence < 70%

Constraint 2: Data Quality Issues

The invoice data is intentionally messy:

- OCR errors in scanned invoices (e.g., "O" vs "0", "I" vs "1")
- Missing fields that need to be inferred
- Inconsistent date formats
- Truncated GSTIN numbers
- Partial vendor names

Constraint 3: Temporal Validity

GST rates have changed over time. Your agent must:

- Apply rules based on invoice date, not processing date
- Handle transition period invoices (March-April boundary)
- Account for retrospective amendments

Constraint 4: Stateful Validation

Some checks require context across multiple invoices:

- Aggregate TDS thresholds
- Duplicate detection across the batch
- Sequential invoice number analysis
- Credit note linkage to original invoice

Constraint 5: The "Historical Trap"

The `historical_decisions.jsonl` contains past decisions - BUT 15% ARE INCORRECT.

Your agent should NOT blindly learn from this data. You must:

- Identify and flag suspicious historical decisions
- Build validation logic from regulations, not patterns
- Document when you deviate from historical precedent

Required Architecture

Your solution MUST demonstrate the following:

1. Multi-Agent Design

Design at least 4 specialized agents:

- **Extractor Agent:** Parse invoices from various formats
- **Validator Agent:** Execute compliance checks
- **Resolver Agent:** Handle conflicts and ambiguities
- **Reporter Agent:** Generate actionable reports

2. Tool Integration

Your agents must use external tools:

- GST portal API simulation (provided mock)
- TDS rate calculator
- Document parser (OCR for images)
- Database for state management

3. Human-in-the-Loop

Implement escalation workflows for:

- Low confidence decisions
- High-value invoices (> ₹10 lakhs)
- First-time vendors
- Regulatory ambiguity

4. Observability

Your system must log:

- Agent reasoning chains
- Tool calls and responses
- Decision confidence scores
- Processing time per invoice

Deliverables

Day 1-2: Foundation

- ☐ System architecture document
- ☐ Agent definitions with roles and tools
- ☐ Data extraction pipeline
- ☐ Basic validation rules (Category A & C)

Day 3: Core Logic

- ☐ GST validation agent (Category B)
- ☐ TDS validation agent (Category D)
- ☐ Conflict resolution logic
- ☐ Integration testing

Day 4: Polish & Edge Cases

- ☐ Policy validation (Category E)
- ☐ Edge case handling
- ☐ Report generation
- ☐ Documentation

Evaluation Criteria

Criterion	Weight	What We're Looking For
Accuracy	30%	Correct compliance decisions on test set
Architecture	25%	Clean agent design, separation of concerns
Edge Case Handling	20%	Graceful handling of messy data
Reasoning Quality	15%	Explainable decisions, proper confidence scoring
Code Quality	10%	Clean, documented, testable code

Accuracy Benchmarks

- **Pass:** > 75% accuracy on standard cases
- **Good:** > 85% accuracy + handles 50% edge cases
- **Excellent:** > 90% accuracy + handles 80% edge cases + proper escalation

Anti-Patterns (Instant Disqualification)

The following will result in immediate disqualification:

6. **Hardcoded decisions** - Mapping invoice IDs to expected results
7. **Single LLM call** - Dumping all data into one prompt
8. **No error handling** - Crashing on malformed data
9. **Ignoring confidence** - Treating all decisions as 100% certain
10. **Copy-paste from historical** - Replicating past decisions without validation

Hints

11. Start with the data - understand the schemas before coding
12. The GST portal mock returns realistic errors - handle them
13. Invoice INV-2024-0847 is a famous edge case in compliance circles
14. Not everything that looks like a GSTIN is valid
15. The company_policy.yaml has a hidden approval matrix
16. Check the fiscal year carefully for TDS thresholds

Support

You may use any LLM provider (OpenAI, Anthropic, local models)

You may use any agent framework (LangChain, AutoGen, CrewAI, custom)

You may NOT ask AI assistants to solve the entire challenge

You MAY ask for clarification on specific regulations

Questions about regulations: dt.ai.coe@datamatics.com

Technical issues: dt.ai.coe@datamatics.com

Good luck! Remember: The goal is not just to pass - it's to build something you'd trust with real money.